

Membrane Air Separation

Pre-Lab Plan
Unit Operations Lab #3
10/3/19
Group #2
Thursday
AM section



Group Members

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1. Experimental Objective

Membrane permeation is a separation process that utilizes the laws of mass transfer and the selective transport through a permeable film. The purpose of the Membrane Separation lab, is to calculate the overall Transport Coefficients of K_{O_2} and K_{N_2} , and determine the relationship with the operating pressure. We can accomplish this using Fick's Law of Diffusion which states that Flux is equal to the diffusivity constants times the concentration gradient divided by the length.

2. Experimental

2.1 Apparatus and Description

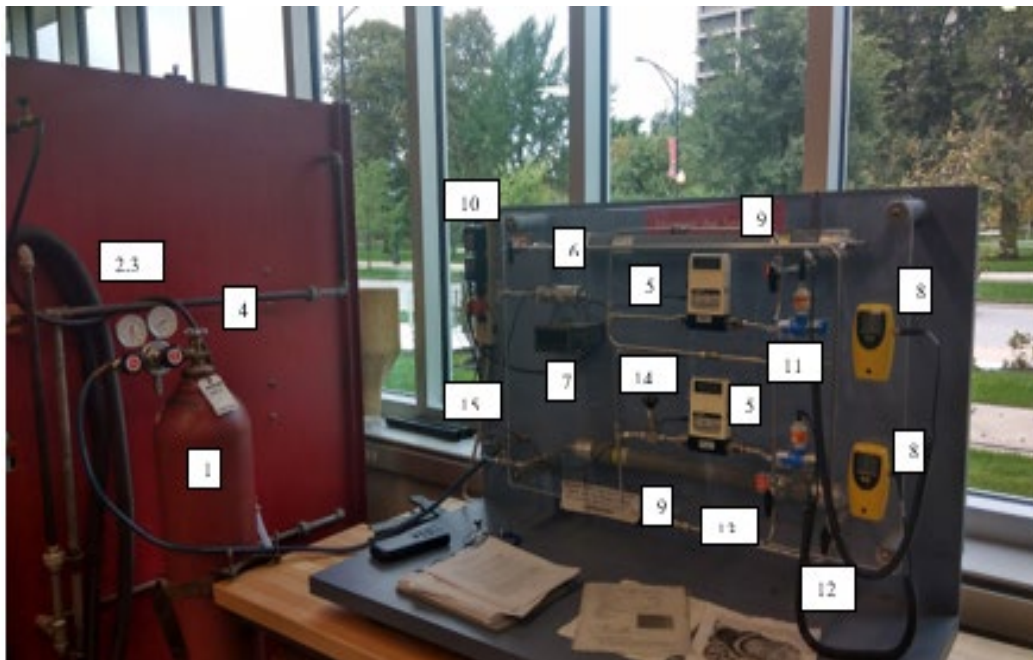


Figure 1: Picture of Membrane Air Separation Apparatus.

Description:

No.	Equipment Part	Use
1.	Air Cylinder	Holds compressed air
2.	Pressure Gauge I (Tank)	Measure pressure outside the tank
3.	Pressure Gauge II (Tank)	Measure pressure inside the tank
4.	Tank Knob	Supplies air into the system
5.	Air flow meter	Measures air flow through the system
6.	Pressure Transducer	Converts pressure into an analog electric signal

7.	Oxygen sensor	Measures the amount of oxygen present in the gas
8.	Digital Oxygen Percent Readers	Output percent Oxygen
9.	Separators (Top and bottom)	Permeates gas through membrane, separating oxygen
10.	Valve A	Controls flow into separators
11.	Valve B	Controls flow into 5
12.	Valve C	Controls flow into 5
13.	Valve D	Controls flow into separators
14.	Turn valve	Controls the direction of gas flow inside piping
15.	Stainless steel tubing	Moves gas throughout the system

2.2 Material/Supplies and Description

This experiment is unique in the fact that apart from the equipment is just compressed air.

Material (or supplies)	Description
Compressed Air	Used as the feed to the membrane air separation apparatus

2.3 Experimental Plan (10 pts)

- **Experimental procedure**

- Open the valve dry compressed cylinder.
 - Set the pressure regulator located on the cylinder to 140 psig.
 - Operation up to 100 psig will be allowed because of this.
- The oxygen analyzers, which are powered by batteries, have to be turned on.
 - Allow the analyzer to reach a steady reading.
 - This will take 10 minutes.
 - The analyzers can be calibrated by carefully removing the gray transducer cell from the aluminum holder and wave the transducer in the air carefully.
 - If the oxygen indicated is not close to 20.9%, read the oxygen sensor calibration procedure.
 - This can be found on the lab bench.
- The mass flow meter must be plugged into the power strip located on the left head side of the apparatus.
 - It only requires a small power supply.
- The pressure transducer display must be plugged to the same power strip.
 - The same power strip is the same one the mass flow meter is plugged into.
- The power strip is equipped with a yellow ground fault circuit interrupter (GFCI).
 - This automatically opens when ground fault is detected.
 - After the ground fault is fixed.
 - Press the reset button to reset the GFCI.
 - The strip is equipped with a 15 ampere overload protector.
- Parallel configuration
 - For the parallel configuration, the setup of the valves should be as follows.
 - Valve A -Left
 - Valve B - Up to sample module 2 shell
 - Down position is to vent.
 - Valve C - Down
 - Valve D - Up to sample module 1 shell.
 - Down position is to vent.
 - The configuration can be easily verified by looking at the piping.
 - The piping is easily visible.
 - Conduct 15 runs in this configuration.
 - Cover a range of operating pressures and flow rates.
 - Time and experience will determine the numbers runs conducted in each corresponding condition.

- Constant flow rates and oxygen levels indicate are noticed. The experimental run can be stopped.
 - This only takes a few minutes.
- Note
 - Changes in the needle valve setting will cause changes in the tube-side flow rate. The feed pressure will slightly be affected, and the pressure regulator must be readjusted.
- Typical set of runs for series or parallel operation.

Operating Tube-Side		
Run number	Pressure (psig)	Flow rate (mL/min)
1	20	500
2	20	1000
3	20	2000
4	20	4000
5	20	8000
6	40	1000
7	40	2000
8	40	4000
9	40	8000
10	80	1000
11	80	2000
12	80	4000
13	80	8000

- A higher flow rate on the tube-side maximizes the oxygen level in the shell-side effluent.
 - This will happen because the tube-side gas will be close to air in oxygen level.
- A lower flow rate on the tube-side minimizes the oxygen level of the tube-side effluent.
 - This is more common if the operation is configured in series.

Series configuration

The valves should be set as follows.

- Valve A - Right
- Valve B - Up to sample module 2 shell
 - Down to vent

- Valve C - up
- Valve D - up to sample module 1 shell
 - Down to vent
- And repeat the same procedure for series.

3. Project Safety Evaluation

 **Project Title:** Membrane Air Separation _____
Date: October 19, 2019

Hazards	Yes/No	Explanation
Potential electrical hazards?	Yes	Large power cord. Make sure that all connections are secure and grounded.
Potential mechanical hazards?	No 	N/A
Potential pressure hazards?	Yes	Gas tank is pressurized and although the piping we must still be careful to slowly pressurize the system. 
Potential temperature hazards?	No	N/A
Hazardous/toxic chemicals involved?	No	N/A
Gloves required?	Yes	Latex gloves
MSDS reviewed by all team members? 	Yes	Yes, Oxygen
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	30C	Slightly above room Temperature
Pressure (psia)	214 	Pressure tank is used

<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
<i>Exit Valve</i>	<i>Yes</i>	Valves exists for emergency outlet for over pressurized gas

I have read relevant background material for the Unit Operations Laboratory entitled: “Binary distillation” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Eduardo Salgado

Signature of team member

10/3/2019

Date

Diego Martinez

Signature of team member

10/3/2019

Date

Yusuf O Sorunke

Signature of team

10/3/2019

Date